

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

IV.Madhulika(192312620) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V. Madhulika

Signature of the student:

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?
(5 Marks)

Answer:

Since Box B is twice as likely as A or C, let $P(A) = P(C) = x$ and $P(B) = 2x$, so $4x = 1$:

- $P(A) = 1/4$, $P(B) = 1/2$, $P(C) = 1/4$

Likelihood of drawing blue from each box:

- $P(\text{Blue}|A) = 4/12 = 1/3$
- $P(\text{Blue}|B) = 7/15$
- $P(\text{Blue}|C) = 2/15$

$$P(\text{Blue}) = (1/4)(1/3) + (1/2)(7/15) + (1/4)(2/15) = 1/12 + 7/30 + 1/30 = 21/60 = 7/20$$

$$P(C|\text{Blue}) = [P(C) \cdot P(\text{Blue}|C)] / P(\text{Blue}) = (1/30) / (7/20) = 2/21 \approx 0.0952$$

$$\text{Result: } P(\text{Box C} | \text{Blue}) = 2/21 \approx 9.52\%$$

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?
(5 Marks)

Answer:

Priors (same reasoning as Q1): $P(A) = P(C) = 1/4$, $P(B) = 1/2$.

- $P(\text{Blue}|A) = 3/10$
- $P(\text{Blue}|B) = 6/15 = 2/5$
- $P(\text{Blue}|C) = 2/15$

$$P(\text{Blue}) = (1/4)(3/10) + (1/2)(2/5) + (1/4)(2/15) = 9/120 + 24/120 + 4/120 = 37/120$$

$$P(C|\text{Blue}) = (1/4 \times 2/15) / (37/120) = (1/30) / (37/120) = 4/37 \approx 0.1081$$

$$\text{Result: } P(\text{Section C} | \text{Blue}) = 4/37 \approx 10.81\%$$

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.
(5 Marks)

Answer:

Total test samples = 20% of $(1800 \times 2 \text{ classes}) = 720$.

$$\text{FN} = \text{Total} - (\text{TP} + \text{TN} + \text{FP}) = 720 - 430 = 290$$

Metric	Formula	Calculation	Result
Accuracy	$(\text{TP} + \text{TN}) / \text{Total}$	$(210 + 190) / 720$	55.56%
Precision	$\text{TP} / (\text{TP} + \text{FP})$	$210 / 240$	87.5%
Sensitivity (Recall)	$\text{TP} / (\text{TP} + \text{FN})$	$210 / 500$	42%
Specificity	$\text{TN} / (\text{TN} + \text{FP})$	$190 / 220$	86.36%

Interpretation: Precision is high (87.5%) but sensitivity is low (42%), meaning the model misses many actual malignant cases (FN = 290). In cancer diagnosis this is risky, since missed positives (false negatives) can delay treatment; the model should be tuned to raise recall even if it costs some precision.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

(5 Marks)

- (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
- (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
- (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
- (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
- i) Identify the epoch at which the model begins to overfit.
- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data

Answer:

- i) Overfitting begins around Epoch 4–5: validation loss is lowest at Epoch 3 (2.0), ticks up at Epoch 4 (2.1), and rises every epoch after (2.3, 2.6, 2.9, 3.2) while training loss keeps falling — Epoch 4 is the best generalization point, Epoch 5 onward is clearly overfitting.
- ii) Two techniques: Early stopping (stop training near Epoch 4); Regularization (L1/L2) or Dropout to prevent memorizing training data.
- iii) As overfitting progresses, the model memorizes training-set noise instead of generalizable patterns. Training loss keeps dropping but validation loss rises, widening the train–validation gap — so the model performs well on training data but poorly on unseen/real-world data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs:

(5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
- i) Identify the epoch at which the model starts overfitting.
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance.

Answer:

- i) Validation accuracy peaks at Epoch 5 (73%) then falls every epoch after (72%, 70%, 68%) while training accuracy keeps rising — overfitting begins around Epoch 5–6.
- ii) The model increasingly fits training-set-specific patterns/noise rather than generalizable features, so training accuracy keeps climbing while performance on the unseen validation set degrades.
- iii) Two methods: Early stopping at the peak validation epoch (Epoch 5); Regularization/dropout or more training data via augmentation to improve generalization.

6. A machine learning model shows the following loss values during training:

(5 Marks)

- (a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
- (b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
- (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
- (d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
- (e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
- (f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
- (g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
- (h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
- i) At which epoch does the model achieve optimal generalization?
- ii) Identify signs of overfitting from the data.
- iii) Recommend two regularization techniques and explain their impact.

Answer:

- i) Epoch 4 gives the lowest validation loss (2.0) with a small train–validation gap — the best generalization point.
- ii) From Epoch 5 onward, training loss keeps falling (1.9 → 1.4) while validation loss keeps rising (2.2 → 3.1) — a widening gap that signals overfitting.
- iii) L2 Regularization (weight decay) — penalizes large weights, reducing model complexity. Dropout — randomly deactivates neurons during training, preventing over-reliance on specific neurons and improving generalization.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).
(5 Marks)

- a) Front identified as Front: 119
- b) Front identifies as Back: 06
- c) Back identified as Back: 120
- d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer:

Taking 'Front' as the positive class: TP = 119, FN = 6, TN = 120, FP = 5, Total = 250.

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/Total$	$(119+120)/250$	95.6%
Precision	$TP/(TP+FP)$	$119/124$	95.97%
Sensitivity	$TP/(TP+FN)$	$119/125$	95.2%
Specificity	$TN/(TN+FP)$	$120/125$	96%
F1-Score	$2 \cdot (P \cdot R)/(P+R)$	$2(0.9597 \times 0.952)/(0.9597+0.952)$	95.58%

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy):
(5 Marks)

- a) Diseased identified as Diseased: 138
- b) Diseased identified as Healthy: 12
- c) Healthy identified as Healthy: 135
- d) Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score**

Answer:

Taking 'Diseased' as the positive class: TP = 138, FN = 12, TN = 135, FP = 15, Total = 300.

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/Total$	$(138+135)/300$	91%
Precision	$TP/(TP+FP)$	$138/153$	90.2%
Sensitivity	$TP/(TP+FN)$	$138/150$	92%
Specificity	$TN/(TN+FP)$	$135/150$	90%
F1-Score	$2 \cdot (P \cdot R)/(P+R)$	$2(0.902 \times 0.92)/(0.902+0.92)$	91.09%

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam):
(5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score**

Answer:

Taking 'Spam' as the positive class: TP = 180, FN = 20, TN = 170, FP = 30, Total = 400.

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/Total$	$(180+170)/400$	87.5%
Precision	$TP/(TP+FP)$	$180/210$	85.71%
Sensitivity	$TP/(TP+FN)$	$180/200$	90%
Specificity	$TN/(TN+FP)$	$170/200$	85%
F1-Score	$2 \cdot (P \cdot R)/(P+R)$	$2(0.8571 \times 0.9)/(0.8571+0.9)$	87.8%

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non defective):

(5 Marks)

- Defective identified as Defective: 230
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

Answer:

Taking 'Defective' as the positive class: TP = 230, FN = 20, TN = 225, FP = 25, Total = 500.

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/Total$	$(230+225)/500$	91%
Precision	$TP/(TP+FP)$	$230/255$	90.2%
Sensitivity	$TP/(TP+FN)$	$230/250$	92%
Specificity	$TN/(TN+FP)$	$225/250$	90%
F1-Score	$2 \cdot (P \cdot R)/(P+R)$	$2(0.902 \times 0.92)/(0.902+0.92)$	91.09%

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.

Answer:

Find the probability the bottle came from Shop X (and from Shop Y).

Priors: $P(Y) = 2 \cdot P(X)$, $P(X)+P(Y) = 1 \rightarrow P(X) = 1/3$, $P(Y) = 2/3$.

- $P(\text{Adulterated}|X) = 30/50 = 3/5$
- $P(\text{Adulterated}|Y) = 20/60 = 1/3$

$$P(\text{Adulterated}) = (1/3)(3/5) + (2/3)(1/3) = 9/45 + 10/45 = 19/45$$

$$P(X|\text{Adulterated}) = (1/5) / (19/45) = 9/19 \approx 0.4737$$

$$P(Y|\text{Adulterated}) = 1 - 9/19 = 10/19 \approx 0.5263$$

Result: $P(\text{Shop X} | \text{Adulterated}) = 9/19 \approx 47.37\%$, $P(\text{Shop Y} | \text{Adulterated}) = 10/19 \approx 52.63\%$

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

Answer:

Find the probability the item came from Section A (and from Section B).

Priors: $P(A) = 3 \cdot P(B)$, $P(A) + P(B) = 1 \rightarrow P(A) = 3/4$, $P(B) = 1/4$.

- $P(\text{Defective}|A) = 15/50 = 3/10$

- $P(\text{Defective}|B) = 25/50 = 1/2$

$$P(\text{Defective}) = (3/4)(3/10) + (1/4)(1/2) = 9/40 + 5/40 = 14/40 = 7/20$$

$$P(A|\text{Defective}) = (9/40) / (7/20) = 9/14 \approx 0.6429$$

$$P(B|\text{Defective}) = 1 - 9/14 = 5/14 \approx 0.3571$$

$$\text{Result: } P(\text{Section A} | \text{Defective}) = 9/14 \approx 64.29\%, \quad P(\text{Section B} | \text{Defective}) = 5/14 \approx 35.71\%$$

13. In a binary classification problem using a deep learning model with a K Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks)

Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Answer:

Test samples per class = 20% of 1000 = 200 (actual positives = actual negatives = 200).

$$FN = 200 - 120 = 80 \quad FP = 200 - 150 = 50$$

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/\text{Total}$	$(120+150)/400$	67.5%
Precision	$TP/(TP+FP)$	$120/170$	70.59%
Sensitivity	$TP/(TP+FN)$	$120/200$	60%
Specificity	$TN/(TN+FP)$	$150/200$	75%

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Answer:

Test samples per class = 25% of 1200 = 300 (actual positives = actual negatives = 300).

$$FN = 300 - 180 = 120 \quad FP = 300 - 160 = 140$$

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/\text{Total}$	$(180+160)/600$	56.67%
Precision	$TP/(TP+FP)$	$180/320$	56.25%
Sensitivity	$TP/(TP+FN)$	$180/300$	60%
Specificity	$TN/(TN+FP)$	$160/300$	53.33%

15. A deep learning model is trained for a classification task, and the following loss values are observed:

(5 Marks)

- i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
- ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2
- iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
- iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
- v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
- vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
- vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
- viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

- a. At which epoch does the model achieve the best generalization performance?
- b. At which point does overfitting begin? Justify your answer.
- c. Suggest two techniques to reduce overfitting and briefly explain how they help.

Answer:

- a) Epoch 4 gives the lowest validation loss (1.8), close to the training loss (1.9) — the best generalization point.
- b) Overfitting begins at Epoch 5: from there, training loss keeps falling (1.7 → 1.2) while validation loss keeps rising (1.9 → 2.7), showing the model is memorizing training data rather than generalizing.
- c) Early stopping — stop training at Epoch 4, before validation performance degrades. Dropout / L2 Regularization — dropout randomly disables neurons during training and L2 penalizes large weights, both encouraging simpler, more generalizable models.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand